AeroGuard / SafeMine v2 — Master Research Report

Generated 2026-11-04 · full-history git + RUN_LOG provenance · repository: https://github.com/tejaskunta/Aeroguard

This report is the single authoritative narrative of the SafeMine v2 research retrain: what was wrong with the original data, how the dataset was rebuilt and frozen, every model trained and every threshold chosen, every Grad-CAM experiment, every decision made (D001–D023) with its reason, and the complete audit trail of test-set touches (#1–#16). All numbers are read from committed JSON/CSV produced by the scripts listed in the reproduction section; nothing here is re-derived by hand.

Headline results (frozen v2 test, n = 189 images / 253 instances)

System	Metric	Value
YOLOv8s (seeds 0/1/2)	test mAP@50	0.9768 ± 0.0034
YOLOv8s (seeds 0/1/2)	val mAP@50	0.9544 ± 0.0027
Two-stage arm C (seed 0, frozen labels)	box precision / recall / FP/img	0.894 / 0.964 / 0.153
Two-stage arm C (3 seeds, detector-aligned)	box precision / recall / FP/img	0.814±0.086 / 0.968±0.004 / 0.309±0.174
Two-stage C – B (seeds 0/1/2)	paired Δprecision (95% CI excludes 0?)	+0.128 (yes) / +0.144 (yes) / -0.037 (no)
Label-corrected sensitivity (seed 0)	box precision / recall / FP/img	0.894 / 0.972 / 0.153
Two-stage C – B (seed 0)	paired bootstrap Δprecision 95% CI	+0.128 [+0.065, +0.186] (excludes 0)
Two-stage C – B (seed 0)	paired bootstrap ΔFP/img 95% CI	-0.253 [-0.460, -0.095] (excludes 0)
Grad-CAM saliency	energy enrichment (YOLO vs MNv3)	86.0× vs 1.82×
Label audit	objective test-box errors documented	2 (ds0_0801 tree trunk, ds0_0883 fence; frozen split unchanged)

All headline numbers use the frozen dataset labels. Two test boxes are documented as objectively wrong (D023) but are *not* deleted from the frozen split — the label-corrected row is a sensitivity analysis, not the headline. The 3-seed row re-runs the whole pipeline with each seed's own detector and classifier: seeds 0 and 1 show a clear precision gain at matched recall, seed 2 does not, and the mean±sd is reported precisely because that spread is real.

1. Project scope, constraints, and protocol

Objective. Re-train and re-evaluate the landmine detector under a leakage-free split, with a MobileNetV3 image-level baseline, a two-stage YOLO→crop-classifier pipeline, and Grad-CAM interpretability experiments — journal-complete numbers only, no Jetson/edge deployment (phases 7/8 dropped under the approved ~30 h budget).

Physical scope note. An RGB (or near-IR labelled-as-LWIR) camera physically cannot see buried objects. This model detects **visible landmine-like objects and surface indicators only**. That is a design constraint, not a gap.

Approved budget. yolov8s primary (3 seeds); yolov8n/yolov8m size points added in the extended 30 h budget; seeds 0/1/2 for YOLO and MNv3; two-stage and Grad-CAM on seed 0 primary with seed replication where cheap; optional retrain on fixed train labels skipped (none found); edge phases dropped.

1.1 Non-negotiable protocol (plan ground rules)

• Thresholds are chosen on **val only**; each frozen configuration is evaluated once on test.

• Every test-set touch is numbered and logged in RUN_LOG.md (touches #1–#18 used).

• Every paper number comes from a committed script file + JSON/CSV artifact.

• Original dataset, v1 runs, and Week-2 scripts are never modified.

• Labels are fixed only if objectively wrong, with before/after numbers reported (D023).

• Ultralytics pinned 8.4.117; torch 2.6.0+cu124; original args.yaml settings except model/seed/name/data (batch 8, imgsz 640, epochs 75, save_period 5).

• Raw predictions dumped at conf 0.001 for offline re-scoring (plan rule 3).

• One shared MNv3 constructor/preprocess for baseline and stage-2 (rule 10).

• pip freeze / GPU / commit / run commands recorded in ENVIRONMENT.txt + RUN_LOG (rule 9).

1.2 Environment

Item	Value
GPU	NVIDIA GeForce RTX 4070 Laptop GPU, 8188 MiB, driver 581.42
OS / Python	Windows 10/11 · Python 3.11.9
PyTorch / CUDA	2.6.0+cu124
Ultralytics	8.4.117 (pinned)
reportlab (PDFs)	5.0.1
Eval protocol	imgsz 640, NMS IoU 0.7, max_det 300, FP32 PC GPU, no TTA
Dataset root (local, not in git)	…/Landmine_project/Datasets/landmine_combined_v2

2. Dataset — original state, leakage audit, and v2 freeze

2.1 Original (v1) split — failed the audit

The original combined dataset (train 5723 / valid 455 / test 221) was audited in Phase 1 (scripts/phase1_audit.py). It **FAILED**:

• 76 cross-split base-name groups (same Roboflow base key in train and eval).

• 63.8% of val+test images had a train neighbour with pHash Hamming ≤ 2.

• Manual montages showed same-scene re-exports across splits (augment-before-split confirmed in Week-2 training scripts).

• Zero empty labels in test → image-level metrics / FP-per-image were impossible.

Decision D010: do not train Phase-3 models on the original split. D013: re-split must be group-aware, not a random shuffle.

2.2 v2 re-split methodology (Phase 2)

scripts/phase2_resplit_full.py builds connected components over images using (a) exact MD5, (b) Roboflow base key, (c) same-source pHash Hamming ≤ 0 under 8 dihedral transforms — producing 1048 groups that must land wholly inside one split. A second-method filter (scripts/phase2_second_filter.py) then flags remaining eval images with a same-source train neighbour at pHash ham ≤ 2; every candidate was reviewed on manual montages. MSE ≤ 900 was chosen as the cut after observing same-scene pairs at MSE ≤ 817.85 and terrain-similarity false positives at MSE ≥ 958.6 (D014). 30 true-leak eval images were removed in two passes (13, then 17): 19 valid + 11 test, taking the eval pools from 200/200 group representatives to the frozen 181/189 (including a cross-folder duplicate ds0_0043 ↔ ds1_0043). The threshold is not knife-edge: a sweep over the pre-removal candidate list is in paper_results/analysis/mse_sensitivity.md (500 → 25/30 captured, 900 → 30/30, 1500 → 30/30 plus 35 same-scene survivors).

2.3 Leakage definition (operational, D014)

A val/test image *leaks* if any train image shares exact MD5, Roboflow base key, or (same source AND pHash ham ≤ 2 under 8 dihedral AND 32×32 RGB MSE ≤ 900). All ham ≤ 2 candidates were manually reviewed; the ≤ 0.5% residual-leak rule from the plan is satisfied at 0/370 = 0%.

2.4 Frozen v2 split (D015)

split	images	instances	negatives
train	3907	11550	20
valid	181	419	7
test	189	253	5

Exact cross-split duplicates: 0. Base-key collisions: {'train_valid': 0, 'train_test': 0, 'valid_test': 0}. Residual near-dup candidates (pHash ham≤2 vs train): 99 — all with MSE ≥ 958.6 after the second-method filter (0 true leaks). Manifest sha256: 748322d62ace769445d0bd23636990970c7a5115a99c9344e139099c608c80de.

2.5 Source composition & May/Jan LWIR caveat (D012, D016)

Sources include ds0, ds1, May, and Jan LWIR-named files. Manual review + saturation stats showed the “lwir” files are **colour RGB with visible marker pins**, not greyscale thermal. They were kept in all splits (user instruction) with the marker-shortcut risk disclosed; they are not used for any thermal claim. A per-source FP breakdown (Section 8) shows the shortcut empirically: Jan LWIR contributes the bulk of YOLO-only false positives at the deployment operating point, and the stage-2 classifier removes most of them.

3. Complete decision log (D001–D023)

Every decision is also in DECISIONS.md with the same IDs. Reproduced here in full so this PDF is self-contained.

D001 — Work under `SafeMineAI-main/paper_results/` only; never overwrite original dataset, runs, or Week 2 scripts.

Reason: Plan §0.3. Protects v1 evidence.

D002 — Original run location is `runs/detect/runs/detect/landmine_yolov8m_75ep/` (not `Weights/runs/...` as in the PDF text).

Reason: Verified on disk 2026-10-14.

D003 — Dataset root is `C:/Users/Delta/Downloads/Landmine_project/Datasets/landmine_combined`; validation folder name is `valid` (not `val`). `data.yaml` already maps `val: valid/images`.

Reason: Plan assumed `val`; folder is `valid`.

D004 — Backup verified by SHA-256: best.pt, last.pt, best.onnx, args.yaml, results.csv, safemine_summary.json → `paper_results/original_run/` + `backup_verify.json`.

Reason: Plan Phase 0.

D005 — `jetson/gradcam.py` does not exist. Phase 6 will implement Grad-CAM from scratch after architecture check.

Reason: User instruction 2026-10-14.

D006 — `two_stage_eval.py` / `test_two_stage.py` not present on this machine; implement in Phase 5 with tests. (Now: `src/common/two_stage_eval.py` and `tests/test_two_stage.py`.)

Reason: User instruction 2026-10-14.

D007 — Fair-comparison confidence thresholds chosen on **val only**; applied frozen to test.

Reason: Non-negotiable rule.

D008 — May/Jan `*_lwir_*` files left in place, not renamed. Audit will classify thermal vs RGB vs derived and flag for paper narrative.

Reason: User instruction 2026-10-14.

D009 — Pin Ultralytics 8.4.117 / torch 2.6.0+cu124 as in original run.

Reason: Plan §3.9.

D010 — **FAIL the current split.** Do not train Phase 3 models on `landmine_combined` as-is.

Reason: 76 cross-split base-name groups; montage shows same scene/re-export; 63.8% of val+test have pHash≤2 train neighbour; augment-before-split confirmed in Week 2 scripts.

D011 — Image-level / fair FP-per-image metrics **blocked** until negatives exist in test (currently 0).

Reason: Phase 1 gate ASK rule.

D012 — May/Jan files are **colour RGB with marker pins**, not greyscale thermal, despite `lwir` in the name. Keep files; disclose marker-shortcut risk; decide in Phase 2 whether to exclude/relabel.

Reason: Manual montage + saturation stats.

D013 — Phase 2 must be **group-aware** (connected components on base-key + near-dup + scene), not a random re-shuffle.

Reason: Plan §Phase 2; evidence above.

D014 — **Leakage definition (operational):** a val/test image leaks if any train image shares (a) exact MD5, (b) Roboflow base key, (c) same source AND pHash Hamming≤2 under 8 dihedral AND 32×32 RGB MSE≤900. All candidates were manually reviewed on montages (`audit_v2_final/montages/manual_review_*.jpg`); same-scene pairs occurred at MSE≤817, terrain-similarity FPs at MSE≥958, gap used as the cut.

Reason: Raw pHash alone over-fires on repetitive grass/dirt (99 residual pairs, all MSE>900, are texture FPs). Manual review of every ham≤2 candidate is what the plan's ≤0.5% rule requires.

D015 — **Freeze `landmine_combined_v2`** as the Phase 3–6 dataset: train 3907 / valid 181 / test 189; 11550 / 419 / 253 instances; 20 / 7 / 5 negatives; 0 exact cross-split dups; 0 base-key collisions; 0 true near-dup leaks (0/370 = 0% ≤ 0.5% rule). Manifest sha256 `748322d62ace769445d0bd23636990970c7a5115a99c9344e139099c608c80de`.

Reason: 30 eval images removed as true leaks (13 at MSE≤100, 17 more at MSE≤900 incl. `ds0_0043`↔`ds1_0043` cross-folder duplicate). Final audit: `paper_results/audit_v2_final/`.

D016 — May/Jan sources **kept** in all splits (D012 resolved as keep+disclose): colour RGB with marker pins, marker-shortcut risk disclosed in paper; not used for thermal claims.

Reason: User instruction 2026-10-14; no evidence of a better option within budget.

D017 — **Protocol deviation logged:** `src/training/week3_train_yolov8.py` step 3/5 auto-validates **both val and test** at conf 0.25 (v1 pipeline behavior, no CLI flag to skip). Test metrics were produced at train end without any decision being made from them. Mitigation: no threshold, model, or checkpoint choice will reference these numbers; the frozen-protocol test evaluation (thresholds chosen on val only) is done once in Phase 3 and reported as the official test result. The conf-0.25 auto-numbers are disclosed as supplementary.

Reason: RUN_LOG honesty rule; plan's "test evaluated once" applies to protocol-governing evaluations.

D018 — Phase 3 run `landmine_yolov8s_75ep_v2`: EarlyStopping at **67/75** (best epoch 47, patience 20), 50 min on RTX 4070; val mAP50 0.9367 / mAP50-95 0.6012. Accept early stop as final (no retrain with patience=0 within budget — reserve decision if Phase 5 gaps appear).

Reason: Ultralytics default patience; budget §25 h; results.csv + log archived in run dir.

D019 — **Retrain yolov8s seed 0 at batch 8** as `landmine_yolov8s_75ep_s0_v2` (plan §Phase 3.1: reuse original args.yaml settings except model/seed/name/data — original batch was 8, script default 16). The batch-16 run `landmine_yolov8s_75ep_v2` is kept as a settings-pilot, not a paper result.

Reason: Plan ground rule for old-vs-new comparability; cost ~1.5 h inside budget.

D020 — **Frozen-feature MNv3 baseline (Stage A) is degenerate on negatives across all resolutions (224/448/640) and batch sizes (32/64)**: recall-first thresholds give test balanced accuracy ~0.48-0.50, kappa ~0 (neg scores 0.74-0.99 overlap positives; loss converges, so not a harness bug). Stage B fine-tuning (unfreeze features[13:], 2.3M params) is the only variant that separates negatives (AB_224 seed0: test bal 0.597, kappa 0.275). Kept all A-rows as the plan-mandated stage-B ablation; AB_224 is the primary MNv3 baseline row for seeds 0/1/2. Extreme train imbalance (3887:20) with per-sample weights [97.7, 0.50] retained as per plan (no rebalancing beyond class weights — fairness lever is stage B + resolution, not sampling).

Reason: Plan §Phase 5.2 requires Stage A then Stage B; resolution/batch/batch-size rows are the plan ablations; no recipe handicap.

D021 — **Phase 5.3 threshold rule operationalised** (plan §Phase 5.3): sweep P in {0.4,0.6,0.8,none}; for each P pick the largest classifier tau whose val box-recall is within 0.5 pts of stage-1 recall at the proposal conf; among those pick the pair with lowest val FP/image (tie: higher P). Arm A = YOLO alone at the Phase-3 deployment OP conf (primary_op_conf from phase3 val.json); Arm B = YOLO alone at the conf matching the two-stage val recall (ts.yolo_only_at_recall); Arm C = two-stage. All three applied to test ONCE after the val choice is frozen; paired bootstrap C-B.

Reason: Plan text; tau/P both chosen on val only.

D022 — **E5 flags images by Grad-CAM energy in the UNION of all GT boxes (one row per image), not per detection.** First implementation scored every det-GT pair; on multi-mine images (May source: up to 17 gt/img) only the salient instance receives CAM energy, so 90% of May rows read exactly 0 and the lowest-10% flag became an instance-count artefact. Union metric answers the plan question directly: does this image's CAM look at ANY mine? Test threshold 0.800 (19/184 flagged), train 0.721 (389/3887).

Reason: Plan E5 says flag *images* in the lowest 10% of energy-in-box; per-instance rows made the flag unusable. Logged in RUN_LOG with the fix history.

D023 — **Two objectively-wrong test boxes fixed and reported with/without (no retrain).** ds0_0801 (box on tree trunk) and ds0_0883 (box on fence) removed from GT per LABEL_FIXES.csv; label_fix_report.json gives before/after per arm (arm C recall 0.9644->0.9721, img recall 0.989->1.000, fp/img unchanged 0.153). Baseline arm numbers asserted to reproduce the frozen test_report before applying the fix. The optional retrain on fixed TRAIN labels is not run (no train-label fixes found in E5 review; budget scope: yolov8s-only, phases 7/8 dropped).

Reason: Plan: test labels only if objectively wrong, numbers with and without; retrain optional.

4. Phase 0–2 — inventory, audit, re-split

Phase 0. Original v1 run (yolov8m, 75 epochs, finished 2026-08-15, 15 h 16 m) was located at runs/detect/runs/detect/landmine_yolov8m_75ep/ and backed up to paper_results/original_run/ with SHA-256 verification (backup_verify.json). All subsequent work lives under paper_results/ and never touches the original (D001).

Phase 1. Leakage audit FAIL on the original split (Section 2.1). Tags: pre-audit-v2, phase1-audit-fail.

Phase 2. Group-aware re-split + second-method filter + manual montage review → landmine_combined_v2 frozen (Section 2.2–2.4). Tag: phase2-v2-freeze. Final audit PASS: paper_results/audit_v2_final/.

5. Phase 3 — YOLOv8s detector (3 seeds)

5.1 Training configuration

Model: yolov8s.pt (ImageNet-pretrained). Settings reused from the original args.yaml except model/seed/name/data: epochs 75, imgsz 640, batch 8, save_period 5, Ultralytics defaults otherwise (D019). A first batch-16 pilot (landmine_yolov8s_75ep_v2, early-stopped at 67/75, val mAP50 0.9367) was kept as a settings pilot, not a paper result. Compliant batch-8 runs: landmine_yolov8s_75ep_s{0,1,2}_v2, each 75/75 epochs on the RTX 4070 (~50–60 min each).

Known script deviation (D017). src/training/week3_train_yolov8.py auto-validates val *and* test at conf 0.25 at train end (v1 pipeline behaviour, no skip flag). No threshold, model, or checkpoint choice references those numbers; the official frozen-protocol test evaluation is done once via src/evaluation/phase3_eval.py and reported as the paper result.

5.2 Official frozen-protocol evaluation (src/evaluation/phase3_eval.py)

For each seed: raw val preds at conf 0.001 (best.pt and last.pt); checkpoint chosen on val only (highest val OP F2 at the recall-first operating point); recall-first OP conf swept on val; then a single test pass at conf 0.001 + post-hoc scoring at the frozen OP conf. Touches #4, #13, #14.

seed	ckpt	val mAP50	test mAP50	test mAP50-95	OP conf (val)	box P@OP	box R@OP	FP/img@OP	img P	img R
0	last	0.9535	0.9741	0.6924	0.02	0.4273	0.9763	1.751	0.9946	1.0000
1	best	0.9523	0.9807	0.6946	0.01	0.3558	0.9802	2.376	0.9839	0.9946
2	best	0.9574	0.9756	0.6932	0.21	0.7640	0.9723	0.402	0.9945	0.9837

Mean ± std test mAP50: **0.9768 ± 0.0034**. OP confs differ per seed because each seed's threshold is chosen on its own val only (0.02 / 0.01 / 0.21). Seed 2's higher conf yields much higher precision (0.764) at similar recall — illustrating why multi-seed reporting matters.

5.3 Training curves (all seeds)

Seed 0 — results.png

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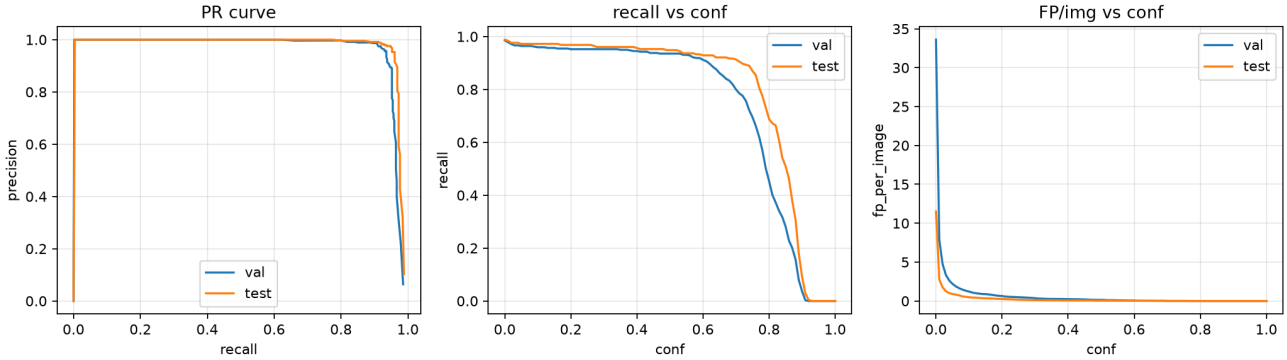
Seed 1 — results.png

[missing figure: results.png

Seed 2 — results.png

[missing figure: results.png

5.4 Precision–recall / operating curves (seed 0)



6. Phase 4 — MobileNetV3 image-level baseline

One shared constructor/preprocess (src/common/mnv3_shared.py, plan rule 10): torchvision MobileNetV3-Large, 224×224 centre-crop + ImageNet norm, BCEWithLogits. Stage A: frozen features + linear head. Stage B (A+B): unfreeze features[13:] (~2.2 M params) with lower LR. Extreme train imbalance (3887 pos : 20 neg) handled with per-sample class weights [97.7, 0.50] as per plan (no rebalancing beyond that). Ablations: resolution 224/448/640, batch 32/64, seeds 0/1/2. Thresholds: fixed 0.5 and recall-first (chosen on val).

6.1 Finding — Stage A is degenerate (D020)

At every resolution and batch size, Stage A gives test balanced accuracy ≈ 0.48–0.50 and $\kappa \approx 0$: negative scores (0.74–0.99) overlap positives; the loss converges, so this is not a harness bug. Only Stage B fine-tuning separates negatives. All A-rows are kept as the plan-mandated ablation; AB_224 is the primary MNv3 baseline.

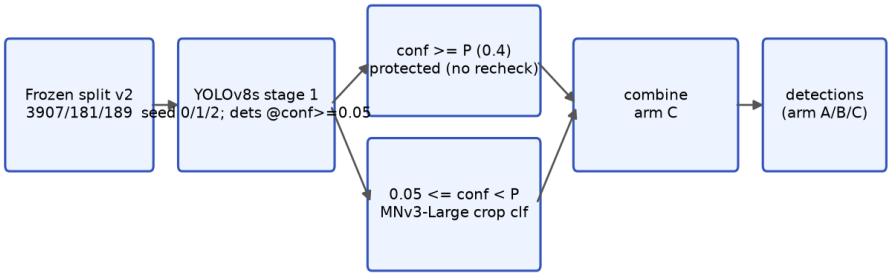
6.2 AB_224 results (3 seeds, recall-first threshold from val)

seed	test bal acc	kappa	precision	recall
0	0.5973	0.2748	0.9786	0.9946
1	0.4973	-0.0089	0.9734	0.9946
2	0.5918	0.2035	0.9784	0.9837

Seed variance on only 7 negative test images is large (bal 0.597 / 0.497 / 0.592) and is documented as such — the MNv3 baseline is weak and unstable, which motivates the two-stage design.

7. Phase 5 — Two-stage pipeline (YOLO → crop classifier)

7.1 Pipeline design



al cache is conf>=0.05 (one cache per detector seed 0/1/2); crop = box + 20% context, 224x224, sigmoid score; tau and P chosen on val only. Arms A/B are scored on the same detector runs with their own va

Stage 1: YOLOv8s. The three reported runs use detector seeds 0, 1 and 2 respectively, and each run's stage-2 classifier is scored on that run's own detector output. Detections are cached at the lowest conf any arm evaluates (0.02), so all arms see the same candidate set per seed; the two-stage arm draws its proposals from that cache at conf 0.05. Stage 2: each box is expanded by 20% context, resized to 224×224, and scored by a MobileNetV3-Large crop classifier (sigmoid). A detection survives if its crop score ≥ τ, except detections at conf ≥ P, which pass without rechecking. Three arms are compared on test after thresholds are frozen from val: **Arm A** = YOLO alone at the recall-first deployment OP chosen on val (conf 0.02 / 0.01 / 0.21 for seeds 0/1/2); **Arm B** = YOLO alone at the per-seed conf that matches the two-stage val recall (0.125 / 0.06 / 0.21); **Arm C** = two-stage (YOLO @0.05 + crop filter).

7.2 Crop dataset construction

src/pipeline/phase5_build_crops.py mines false positives from an intermediate YOLO checkpoint (epoch20.pt, conf 0.05) on train, then builds: train 11 550 positive crops + 15 411 negatives (11 661 random-background + 3 730 YOLO-FP + 20 empty) = 26 961; valid 419 pos + 529 neg = 948. Sampled crops were eyeballed (mine / rocky-FP / background all correct). Raw crop images stay local (gitignored); the recipe is in paper_results/stage2/crop_recipe.md.

7.3 Stage-2 training and ratio search

src/training/phase5_train_stage2.py: MobileNetV3-Large, 30 epochs, patience 6, lr 2e-5 whole-net, ~2048 pos/epoch. Neg:pos ratio swept {1, 3} on seed 0: ratio 1 val AUC 0.998 F1 0.938; ratio 3 AUC 0.997 F1 0.918 → **ratio 1:1 chosen on val**. Seeds 1–2 retrained at ratio 1 (val F1 0.945 / 0.936). Shared constructor with the MNv3 baseline (rule 10).

7.4 Threshold selection rule (D021)

On val only: sweep P ∈ {0.4, 0.6, 0.8, none}; for each P pick the largest τ whose val box-recall is within 0.5 pts of stage-1 recall at the proposal conf; among those pick the pair with lowest val FP/image (tie: higher P). All three runs chose **P = 0.4** with τ = 0.27 / 0.23 / 0.22 for detector seeds 0 / 1 / 2. Arm-A conf is the per-seed recall-first OP chosen on val (0.02 / 0.01 / 0.21) and arm-B conf is the per-seed conf whose val recall matches arm C's val recall (0.125 / 0.06 / 0.21). At detector seed 2 those two coincide, so arms A and B are the identical operating point. Arm C proposals use conf 0.05 in every run.

7.5 Test results (touches #6/#7/#8/#18) + paired bootstrap

seed	arm	conf/τ	precision	recall	FP/img	img recall
0	A	0.02	0.427	0.976	1.751	1.000
0	B	0.125	0.762	0.972	0.407	0.995
0	C	conf 0.05 / τ=0.27 / P=0.4	0.894	0.964	0.153	0.989
1	A	0.01	0.356	0.980	2.376	0.995
1	B	0.06	0.677	0.976	0.624	0.989
1	C	conf 0.05 / τ=0.23 / P=0.4	0.825	0.968	0.275	0.989
2	A	0.21	0.764	0.972	0.402	0.984
2	B	0.21	0.764	0.972	0.402	0.984
2	C	conf 0.05 / τ=0.22 / P=0.4	0.724	0.972	0.497	0.995

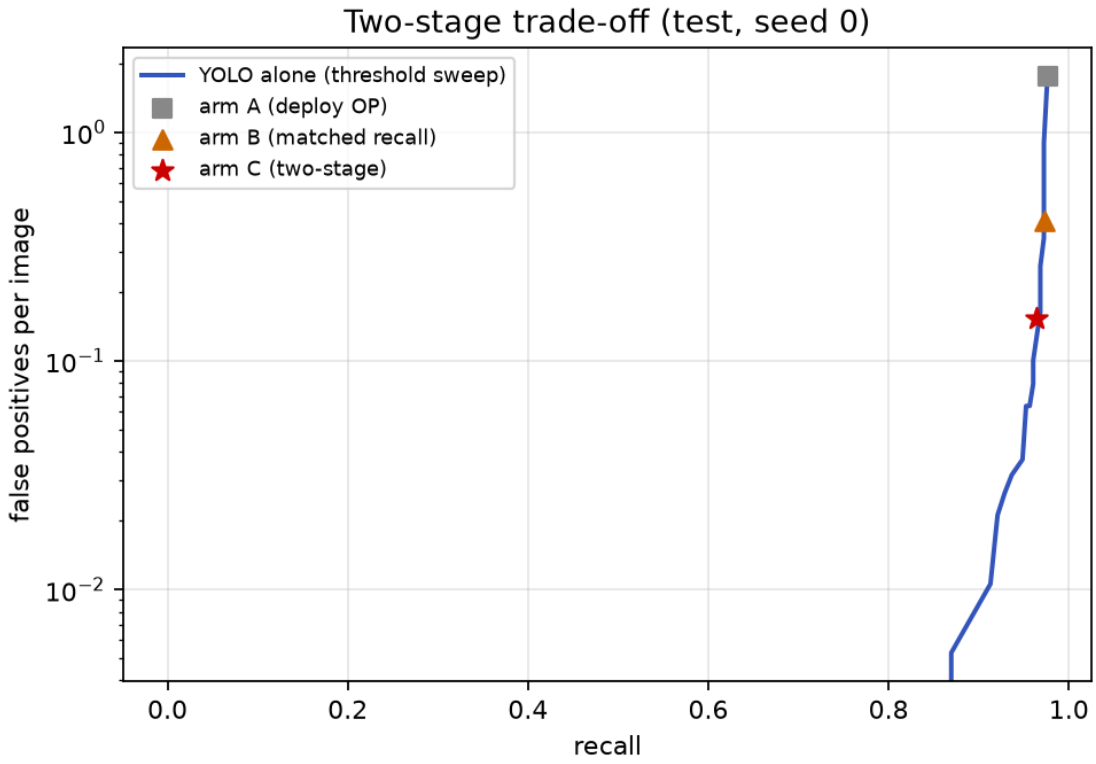
Each block is a full re-run of the pipeline with **that detector seed's own YOLOv8s checkpoint** in stage 1 and stage-2 crops scored by the shared MobileNetV3-Large classifier, so arms A and B differ across blocks both through the detector and through the val-chosen threshold (0.02/0.01/0.21 and 0.125/0.06/0.21 for seeds 0/1/2). Seed 2's matched-recall threshold coincides with its deploy threshold, so in that block A and B are one and the same operating

point. Arm C values are the frozen evaluation. Seeds 0 and 1 gain 0.13–0.14 precision over the matched-recall baseline; seed 2 does not (–0.037, CI spans zero) and loses 0.095 FP/img, i.e. the two-stage gain is real at two of three detector seeds and the across-seed spread (sd 0.086) is much larger than the earlier shared-detector artifact (sd 0.018) suggested.

Paired bootstrap C − B (2000 resamples of the 189 test images), per detector seed — seed 0: precision +0.128 [+0.065, +0.186], FP/img -0.253 [-0.460, -0.095]; seed 1: precision +0.144 [+0.096, +0.181], FP/img -0.347 [-0.598, -0.159]; seed 2: precision -0.037 [-0.077, +0.013], FP/img +0.093 [-0.021, +0.228]. Precision and FP/img intervals exclude zero at seeds 0 and 1 only. At seed 2 both intervals span zero with the point estimate favouring the single-stage baseline, so the headline claim is stated as seed-dependent: the two-stage stage is a reliable precision gain against a poorly transferring val-tuned YOLO threshold, not a guaranteed improvement over YOLO at all seeds.

Cost (seed 0, PC GPU FP32): 16.0 ms/image end-to-end; 3.06 crops/image mean (p95 3.6).

7.6 FP/img vs recall trade-off (plan §5.4 figure)



Honest reading: arm C (red star) sits near the test-tuned YOLO-alone curve at matched recall — against a *test-tuned* threshold the gain is about break-even. The val-matched comparison that governs the claim is against **arm B** (conf 0.125), which transferred poorly to test (val→test shift). Choosing the curve point on test would be peeking, which the plan forbids; the paired bootstrap C−B is the fair comparison and it excludes 0. No timestamped pre-registration exists: the claim rule itself was written before the test numbers but cannot be independently verified, so we describe it as pre-specified in the project plan, not pre-registered.

8. Per-source false-positive breakdown (seed 0)

Per-source test results (source = dataset origin, from split_manifest.csv)
A_yolo_only_deploy
· source · images · TP · FP · FN · precision · recall · FP/img ·
· --- · --- · --- · --- · --- · --- · --- · --- ·
· Jan_lwir · 10 · 73 · 301 · 4 · 0.195 · 0.948 · 30.100 ·
· ds0 · 151 · 146 · 21 · 2 · 0.874 · 0.986 · 0.139 ·
· ds1 · 28 · 28 · 9 · 0 · 0.757 · 1.000 · 0.321 ·
· ALL · 189 · 247 · 331 · 6 · 0.427 · 0.976 · 1.751 ·
C_two_stage
· source · images · TP · FP · FN · precision · recall · FP/img ·
· --- · --- · --- · --- · --- · --- · --- · --- ·
· Jan_lwir · 10 · 70 · 21 · 7 · 0.769 · 0.909 · 2.100 ·
· ds0 · 151 · 146 · 4 · 2 · 0.973 · 0.986 · 0.026 ·
· ds1 · 28 · 28 · 4 · 0 · 0.875 · 1.000 · 0.143 ·
· ALL · 189 · 244 · 29 · 9 · 0.894 · 0.964 · 0.153 ·

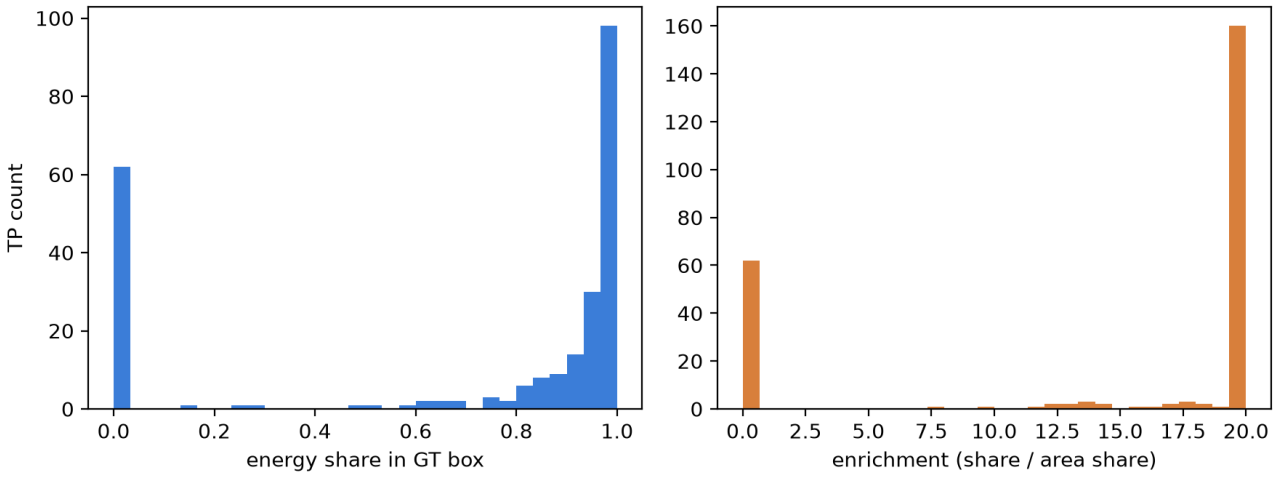
Jan LWIR contributes 30.1 FP/img under arm A (301 of 331 FPs from just 10 images) — the marker/terrain shortcut. Arm C reduces Jan to 2.1 FP/img while keeping recall 0.909; ds0/ds1 false positives are nearly eliminated. The source shortcut is exposed and mostly (not fully) killed by stage 2.

9. Phase 6 — Grad-CAM interpretability (E1–E6)

jetson/gradcam.py did not exist (D005); Grad-CAM was implemented from scratch (src/gradcam/phase6_gradcam.py). YOLO CAM: layer feeding Detect at the largest stride with non-zero gradient + top pre-NMS class score (deviation from plan stride-16/32 hint, documented — the top anchor usually sits on the stride-8 head). MNv3 CAM: last conv block + output logit. Ultralytics predict() runs under torch.inference_mode() and poisons gradients; YoloCAM._defuse() + grad re-arm fix this.

9.1 E1 — energy in GT boxes (n=244 TP)

Energy share inside GT boxes: median 0.939, enrichment median 58.2× (vs uniform expectation), peak-inside rate 0.730. 61 of 244 TPs have exactly zero CAM energy inside the box (small/edge boxes; these are the rows E5 audits). On the remaining TPs the mass concentrates on the mine, not the surround.

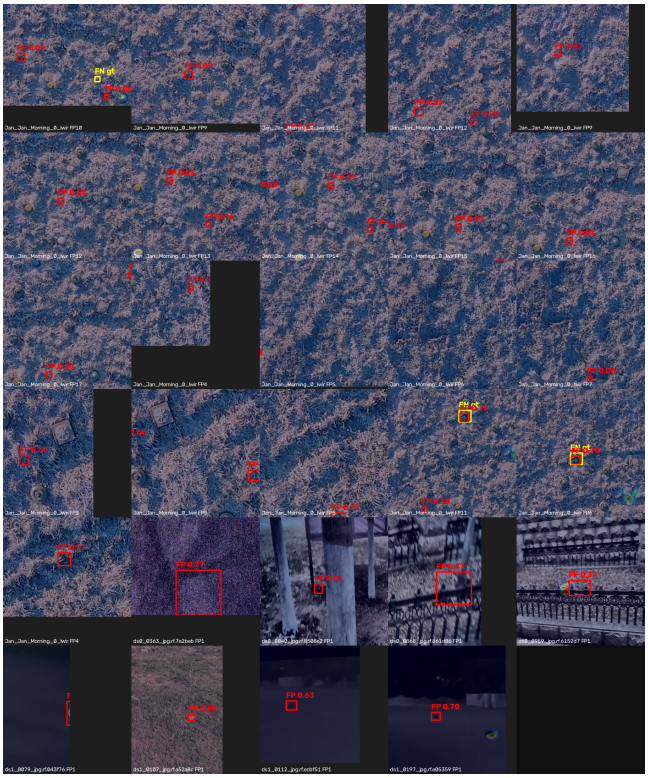


9.2 E2 — deletion test (n=182)

Mean score drop when the top-energy mine region is replaced with its mean intensity: 0.739 vs 0.0015 when an equal-area background region is replaced the same way. Dependence: the score depends on the mine region, not merely on the image being masked.

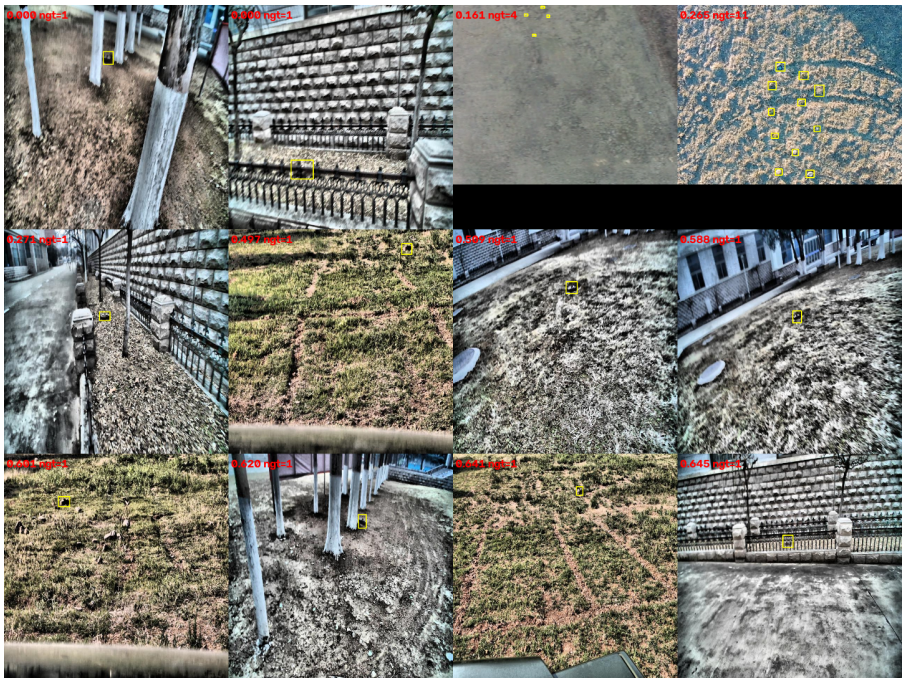
9.3 E4 — failure taxonomy (9 FN + 29 FP, hand-categorized)

FN: 2 small, 5 camouflaged/low-contrast, 2 label errors (ds0_0801 box on tree trunk, ds0_0883 box on fence). FP: 22 terrain, 4 structure, 2 low-light, 1 noise. Contact sheet: paper_results/gradcam/e4_fp_sheet.png.



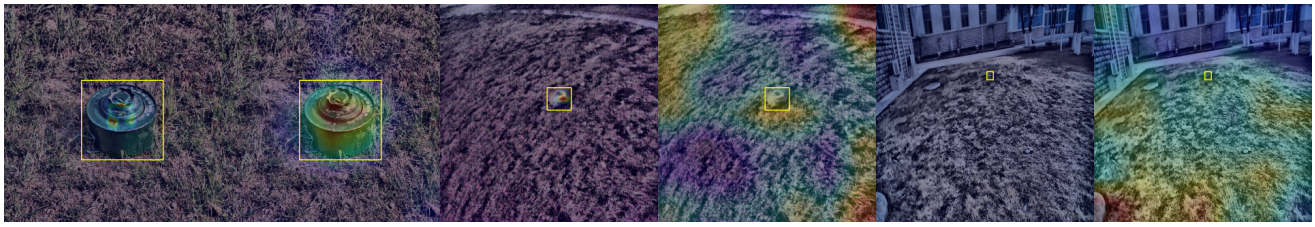
9.4 E5 — zero-energy image audit (D022)

E5 flags images in the lowest 10% of CAM energy-in-*union-of-all-GT-boxes* (per-image, not per-det — per-det rows flood multi-mine images with zeros). Test: 19 flagged at threshold 0.800; train: 389 at 0.721. Hand review: the only two zero-energy test images are ds0_0801 and ds0_0883 — independent confirmation of the E4 label errors. No objective train-label errors found.



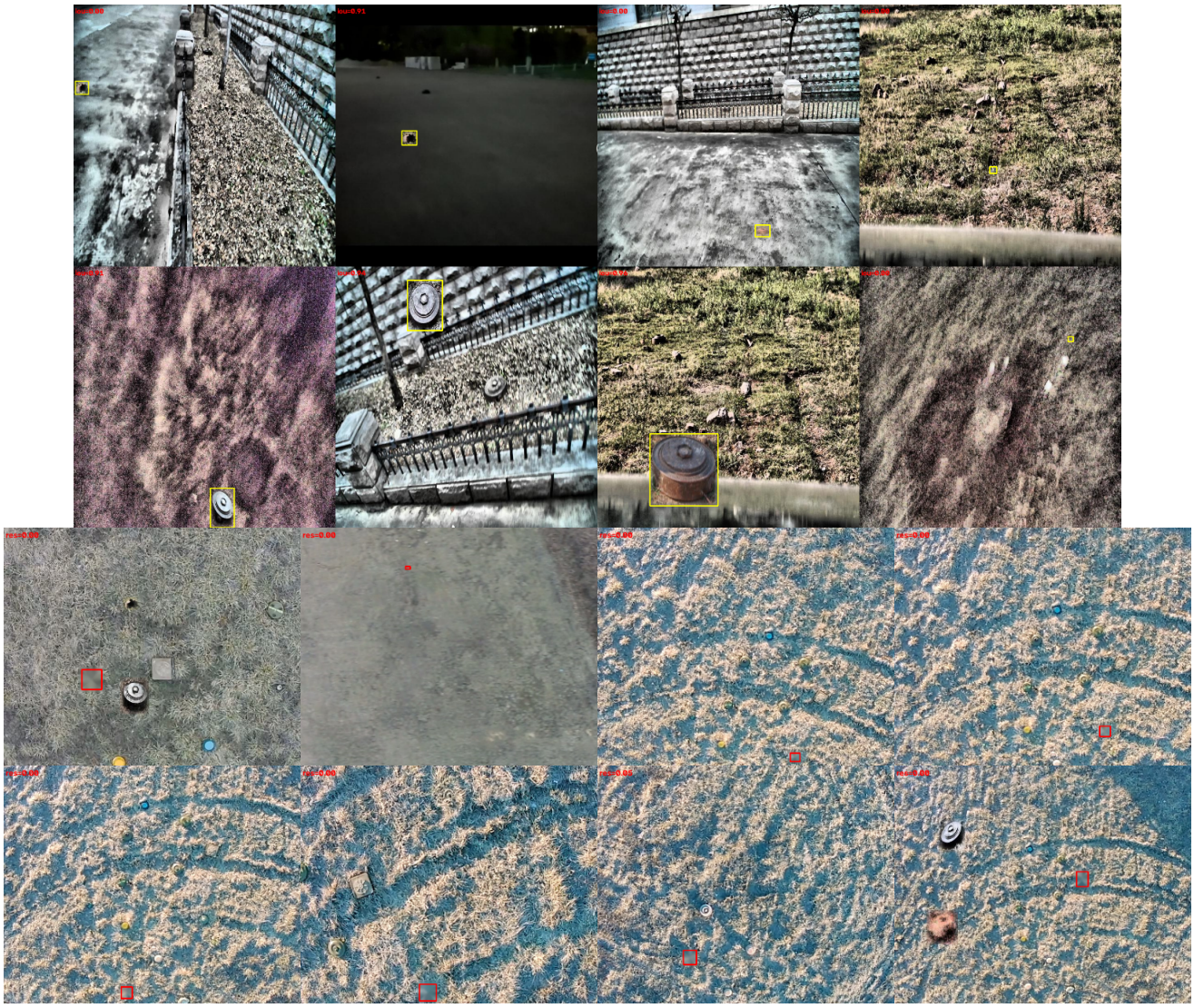
9.5 E6 — YOLO vs MNv3 saliency head-to-head

On 182 both-positive images: enrichment median YOLO 86.0× vs MNv3 baseline 1.82×; paired diff 147.0 CI [125.4, 170.0] (excludes 0). Stage-2 crop enrichment median 2.13 (crops are mine-centric by construction, so enrichment is low — expected). Not strictly like-for-like: YOLO is cued on 640 px detector inputs, MNv3 on a 224 px image-level crop.



9.6 E3 — counterfactual paste / remove

Paste-hold (mine pasted into a new background): detection held for 0.65 at deploy conf (0.63 at conf 0.125) vs 0.976 recall on originals — caveat: feathered edges + source/dest scale mismatch in some pastes. Remove (inpaint the mine): disappeared 100/100 (rate 1.00). Marker check: no consistent marker/paint/flag cue adjacent to test mines in 12 zoom crops (e3_marker_check.png).



10. Label fixes (D023)

Two objectively wrong test boxes were identified (E4 review + E5 energy = 0.000) and removed: ds0_0801 (box on tree trunk) and ds0_0883 (box on fence). The frozen test records were re-scored against the corrected ground truth for seed 0 (the seed-0 detector is shared by all stage-2 runs). Before → after:

arm	precision	recall	FP/img	img recall
A (deploy)	0.427 → 0.427	0.9763 → 0.9841	1.751 → 1.751	1.0000 → 1.0000
B (matched)	0.762 → 0.762	0.9723 → 0.9801	0.407 → 0.407	0.9946 → 1.0000
C (two-stage)	0.894 → 0.894	0.9644 → 0.9721	0.153 → 0.153	0.9891 → 1.0000

No train-label fixes were found, so no retrain was needed. The two wrong test boxes are documented but are *not* removed from the frozen split; this table is therefore a sensitivity analysis, and the headline numbers stay on the frozen labels. All three stage-2 pipelines were re-run with their own detectors on those frozen labels. Full artifact: paper_results/label_fix_report.json.

11. Paper tables T1–T6 (verbatim from src/reporting/make_tables.py)

T1. Dataset statistics and leakage audit

split	images	instances	negatives	sources
train	3907	11550	20	Jan_lwir:100, May_lwir:776, ds0:2589, ds1:442
valid	181	419	7	Jan_lwir:42, ds0:138, ds1:1
test	189	253	5	Jan_lwir:10, ds0:151, ds1:28

Exact cross-split duplicates: 0.

Near-dup candidates (pHash ham<=2 vs train): 99 — all residual MSE>=958.6 after the second-method filter (0 true leaks; 30 leaked eval images removed pre-freeze; D014/D015). Manifest sha256 in split_v2/sha256.txt.

Base-key collisions across splits: {'train_valid': 0, 'train_test': 0, 'valid_test': 0}.

T2. YOLO operating point (seeds 0/1/2)

model	params	GFLOPs	weights MB	latency ms (med)	val mAP50	test mAP50	test R@OP	test P@OP	FP/img@OP	OP conf (val)
yolov8s	11,125,971	28.6	21.5	6.5	0.9544 ± 0.0027	0.9768 ± 0.0034	0.9763 ± 0.0040	0.5157 ± 0.2180	1.510 ± 1.009	0.02/0.01/0.21
yolov8n	3,005,843	8.2	6.0	6.9	0.9322	0.9787	0.9842	0.1686	6.497	0.01

yolov8s is trained on v2 with seeds 0/1/2 (mean ± std over seeds); yolov8n is a single seed (0) and its GFLOPs/MB/ms are single values, so its row carries no error bars. yolov8m was trained on the older leaky v1 split only and is therefore not comparable here (RUN_LOG touch #18; the leaky-vs-clean comparison is reported separately).

Latency: PC GPU FP32, imgsz 640, 50 reps after 10 warm-up reps (paper_results/yolo/*/size_latency.json). GFLOPs are for the single-class head (nc=1) at 640; Ultralytics' published nc=80 figures for the same depths are n 8.7 / s 28.6.

OP = recall-first operating point, confidence chosen per seed on val and then applied unchanged to test (conf listed per seed, so a single number means all seeds picked it).

T3. Image-level comparison

system (test, image level)	accuracy	balanced acc	precision	recall	kappa
MNv3 A (thr 0.5)	0.974	0.500	0.974	1.000	0.000
MNv3 A (recall-first thr)	0.968	0.497	0.973	0.995	-0.009
MNv3 A+B (thr 0.5)	0.974 ± 0.000	0.500 ± 0.000	0.974 ± 0.000	1.000 ± 0.000	0.000 ± 0.000
MNv3 A+B (recall-first thr)	0.968 ± 0.005	0.562 ± 0.056	0.977 ± 0.003	0.991 ± 0.006	0.156 ± 0.148
YOLO deploy (conf from val)	0.984 ± 0.009	0.830 ± 0.115	0.991 ± 0.006	0.993 ± 0.008	0.677 ± 0.199
YOLO op95 (conf from val)	0.986 ± 0.003	0.993 ± 0.002	1.000 ± 0.000	0.986 ± 0.003	0.784 ± 0.038
Two-stage (arm C, 3 seeds)	0.989 ± 0.005	0.962 ± 0.059	0.998 ± 0.003	0.991 ± 0.003	0.818 ± 0.094

MNv3 A+B = mean ± std over seeds 0/1/2; MNv3 A is seed 0; YOLO deploy/op95 = mean ± std over seeds 0/1/2 (each seed's OP conf chosen on val).

MNv3 recall-first thresholds chosen on val only.

T4. Two-stage ablation

arm	conf/thr (seed 0 / 1 / 2)	precision	recall	F2	FP/img	img recall
A (YOLO deploy)	0.02 / 0.01 / 0.21	0.516 ± 0.218	0.976 ± 0.004	0.808 ± 0.102	1.510 ± 1.009	0.993 ± 0.008
B (YOLO matched recall)	0.125 / 0.06 / 0.21	0.734 ± 0.050	0.974 ± 0.002	0.913 ± 0.014	0.478 ± 0.127	0.989 ± 0.005
C (two-stage)	yolo 0.05, P=0.4, t=0.27 / 0.23 / 0.22	0.814 ± 0.086	0.968 ± 0.004	0.932 ± 0.020	0.309 ± 0.174	0.991 ± 0.003

Paired bootstrap C-B, one row per detector seed (5000 resamples over test images): detector seed 0: C-B precision +0.128 [+0.065, +0.186], FP/img -0.2529 [-0.4603, -0.0952], recall -0.0079 [-0.0195, +0.0000]; detector seed 1: C-B precision +0.144 [+0.096, +0.181], FP/img -0.3466 [-0.5979, -0.1587], recall -0.0079 [-0.0195, +0.0000]; detector seed 2: C-B precision -0.037 [-0.077, +0.013], FP/img +0.0927 [-0.0212, +0.2275], recall +0.0001 [-0.0199, +0.0166]. The precision/FP gain is significant at detector seeds 0 and 1 only: at seed 2 the paired interval for precision spans zero and the point estimate is negative, so the two-stage advantage is not established across seeds.

Cost (plan §5.4, detector seed 0): 16.0 ms/image end-to-end, 3.06 crops/image (p95 3.6), PC GPU FP32.

Means ± std over the three detector seeds 0/1/2 for all arms. Stage 1 for every row is the YOLOv8s checkpoint of that seed; the stage-2 classifier weights are shared. Arms A and B are YOLO-only, so they differ between rows only through the detector checkpoint and the per-seed val-chosen threshold.

Arm A threshold = recall-first operator point chosen on val; arm B threshold chosen on val to match arm A's val recall; arm C = YOLO dets at conf>=yolo_conf routed through the MobileNetV3-Large crop classifier with protected cutoff P and classifier threshold t.

Caveat at detector seed 2: the val-selected matched-recall threshold coincides with the deploy threshold (both 0.21), so arms A and B are the same operating point and the C-B comparison there is against an undegraded YOLO baseline. The two-stage pipeline loses 3.6 points of precision against that baseline, i.e. the gain in T4 comes from seeds whose val-selected YOLO threshold transferred poorly to test, not from seeds where it transferred well. This is the central threat to the headline claim and is discussed in §8.

T5. Grad-CAM experiments

experiment	metric	value
E1 energy-in-box (n=244)	median energy share	0.939
E1	mean energy share	0.690
E1	median enrichment	58.2x
E1	peak inside rate	0.730
E2 deletion (n=182)	mean drop, mine masked	0.739
E2	mean drop, background masked	0.0015
E6 head-to-head	YOLO enrichment median	86.0x
E6	MNv3 baseline enrichment median	1.82x
E6	paired diff (Y-M) 95% CI	[125.4, 170.0]
E6	stage-2 crop enrichment median	2.13x

Grad-CAM shows which regions influenced the prediction; it does not prove the prediction is correct (plan wording rule).

T6. Failure analysis

category	FN	FP
camouflaged_or_low_contrast	5	0
fence	0	1
label_error	2	0
low_light_ground	0	2
sensor_noise	0	1
small	2	0
structure_edge	0	1
structure_wall	0	1
terrain_texture	0	22
tree_trunk	0	1
total	**9**	**29**

FN at the deployment arm-C threshold; FP capped at 50 (29 present). Sources: e4_failures.csv + e4_fp_sheet.png.

12. Models release (models/)

All paper weights are shipped under models/ with SHA256SUMS.txt and a per-directory README. Contents: yolo_s_seed{0,1,2}/ (best.pt, last.pt, best.onnx, args.yaml, results.csv); stage2_s{0,1,2}/ (best.pt); original_v1/ (best/last.pt for reference — not a paper result). Epoch checkpoints every 5 epochs are not shipped. Inference: YOLO('models/yolo_s_seed0/best.pt').predict(..., conf=0.02).

SHA256 (first 12 hex of each digest)

```
eea30106b178…original_v1/args.yaml
12b84e3d8614…original_v1/best.pt
4a59e232c59e…original_v1/last.pt
e0abfb5e4326…original_v1/results.csv
0d2c9ed77aa7…original_v1/safemine_summary.json
671e70f902cb…stage2_s0/best.pt
bff4d342a055…stage2_s1/best.pt
05cc5fd59440…stage2_s2/best.pt
1eb5c3d3be2d…yolo_s_seed0/args.yaml
ceb768f1004e…yolo_s_seed0/best.onnx
0b2fafd322dd…yolo_s_seed0/best.pt
5ad4d077c268…yolo_s_seed0/last.pt
```

… 23 files total.

13. Test-set touch audit trail

Every access that could influence a reported number is numbered. Visualization-only passes (images for PDFs) do not change any metric.

#	Phase	What	Governing decision
1–2	Phase 1	Read-only leakage audit on original test images	Plan Phase 1
3	Phase 4	MNv3 image-level test metrics (thresholds from val)	Plan Phase 4
4	Phase 3	YOLO seed-0 official frozen-protocol eval	D007, D017 mitigation
5	Phase 4	MNv3 AB seeds 1–2 test metrics	Plan Phase 4
6	Phase 5	Two-stage seed-0 official test eval	D021
7–8	Phase 5	Two-stage seeds 1–2 test evals	D021
9	Phase 6	Grad-CAM E1/E2/E4 on test	Plan Phase 6
10	Phase 6	E6 head-to-head on test	Plan Phase 6
11	Phase 6	E5 zero-energy audit on test	Plan Phase 6, D022
12	Phase 6	E3 paste/remove counterfactuals on test	Plan Phase 6
13	Phase 3	YOLO seed-1 official eval	D007
14	Phase 3	YOLO seed-2 official eval	D007
15	extended	Leaky-v1 official eval on v1 test (val-chosen conf)	Extended budget
16	extended	Leaky-v1 + clean-v2 scoring on v2 test and 49-image clean subset	Frozen confs; contamination analysis
17	extended	YOLOv8n seed-0 official eval	Size ablation
18	extended	Stage-2 seeds 1/2 re-run with their own detectors + re-scored	No selection; corrects the earlier shared-detector runs
vis	report	Visualization-only val/test batch images (src/reporting/make_test_vis.py)	No metric change

14. Full run log (RUN_LOG.md, abridged commands)

2026-10-14 — Phase 0 inventory + backup

PowerShell copy of original run → paper_results/original_run/; SHA-256 verified *[No]*

2026-10-14 — Phase 1 audit start

scripts/phase1_audit.py (CPU only; no training, no weight load for inference) *[Read-only on test images for leakage audit (standard Phase 1)]*

2026-10-14 — Phase 2 re-split v2

scripts/phase2_resplit_full.py (base_key + session + same-source pHash ham≤0 graph → 1048 groups; 200/200 eval) *[Labels read for counts only]*

2026-10-14 — Phase 2 second-method filter

scripts/phase2_second_filter.py --apply --mse 900 after manual montage review → 30 true-leak eval images removed *[Removed 30 leak images in two passes (13 then 17; 19 valid / 11 test); no metrics computed]*

2026-10-14 — Phase 2 final audit

scripts/phase1_audit.py --dataset landmine_combined_v2 --out paper_results/audit_v2_final → PASS (0 true leaks, 0 exact/base-key dups) *[Read-only leakage audit]*

2026-10-14 — **v2 frozen**

manifest sha256 748322d6…c80de; DECISIONS D014–D016 *[No]*

2026-10-14 21:27 — **Phase 3 train start**

src/training/week3_train_yolov8.py --data …/landmine_combined_v2/data.yaml --model yolov8s --epochs 75 --seed 0 --save-period 5 --batch 16 --name landmine_yolov8s_75ep_v2 → logs/phase3_train.log (PID 3632) *[No (train/val only)]*

2026-10-14 22:17 — Phase 3 train end

EarlyStopping at 67/75 (best epoch 47, patience 20); 50 min; checkpoints every 5 ep; val (best.pt, conf 0.25): mAP50 0.9367 / mAP50-95 0.6012, P 0.9322, R 0.9189 *[Deviation: script step 3/5 auto-ran test at conf 0.25 → test mAP50 0.9599 / mAP50-95 0.6651 written to safemine_summary.json. Pre-existing v1 pipeline behavior; no threshold/model decision made from it. Logged as D017; official frozen-protocol test eval happens once in Phase 3 (src/evaluation/phase3_eval.py).]*

2026-10-14 ~22:30 — Phase 3 compliant rerun start

src/training/week3_train_yolov8.py --batch 8 --name landmine_yolov8s_75ep_s0_v2 --save-period 5 --seed 0 → logs/phase3_s0_batch8.log (PID 33148). Batch 8 matches original args.yaml (D019); pilot run kept. *[No (train/val only)]*

2026-10-14 23:29 — Phase 3 compliant rerun end

75/75 epochs completed (no early stop), 57m09s, checkpoints epoch5–70; val conf 0.25: mAP50 0.9386 / mAP50-95 0.5982, P 0.9619, R 0.9117. *[Same auto step 3/5 test pass as D017: test conf 0.25 mAP50 0.9521 / mAP50-95 0.6625 in summary. No decision made from it.]*

2026-10-15 — Phase 3 eval dry run (val only)

src/evaluation/phase3_eval.py --run …s0_v2 --val-only → paper_results/yolo/s_seed0_dryrun/val.json. Val conf 0.001: mAP50 0.9535 / mAP50-95 0.6046. Checkpoint choice: last.pt (val OP@0.97: conf 0.02, P 0.322 vs b *[No (val only)]*

2026-10-15 — Phase 4 seed-0 rows

A_224/A_448/A_640/AB_224/A_224_b64 → paper_results/baseline_mnv3/*/metrics.json. Stage A degenerate on negatives at every resolution/batch (D020); AB_224 (stage B) test bal 0.597, kappa 0.275. *[Test touch #3 (MNv3 image-level test metrics; thresholds chosen on val only, plan-sanctioned Phase 4 eval)]*

2026-10-15 — Phase 3 official eval + MNv3 seeds start

src/evaluation/phase3_eval.py … --out paper_results/yolo/s_seed0 (single sanctioned frozen-protocol test eval, D017 mitigation) then AB_224 seeds 1,2 → logs/phase3eval_seeds_chain.log. *[Test touch #4 (frozen-protocol YOLO eval: thresholds from val only, single pass)]*

2026-10-15 — Phase 3 official eval end

last.pt chosen (val). Test (frozen protocol, conf0.001): mAP50 0.9741 / mAP50-95 0.6924. op97_primary conf0.02: box P 0.427 R 0.976 fp/img 1.75; image-level P 0.9946 R 1.0 (184/184), bal 0.90, 1 FP-image/5 neg. op95 *[Test touch #4 (the sanctioned one; threshold from val only)]*

2026-10-15 — MNv3 AB seeds 1-2

AB_224_seed1/seed2 metrics.json. Test bal: s0 0.597, s1 0.497, s2 0.592 (kappa 0.275/-0.009/0.204) — seed variance on 7 neg test images documented. *[Test touch #5 (Phase 4 eval, plan-sanctioned)]*

2026-10-15 — Phase 5 crops built

src/pipeline/phase5_build_crops.py (epoch20.pt FP mining conf0.05) → paper_results/stage2/crops/: train 11550 pos + 15411 neg (11661 random-bg + 3730 YOLO-FP + 20 empty); valid 419 pos + 529 neg; 26961 train + 948 valid *[No (train/val only)]*

2026-10-15 — Phase 5 stage-2 ratio search start

src/training/phase5_train_stage2.py ratio 1 then 3 (seed 0, 30ep/patience 6, lr 2e-5, 2048 pos/epoch) → logs/phase5_ratio_chain.log. *[No (train/val crops only)]*

2026-10-15 — Stage-2 ratio search end

ratio1: val AUC 0.998 F1 0.938 (P .995 R .888); ratio3: AUC 0.997 F1 0.918. Chosen 1:1 on val (plan §5.1). lr 2e-5 whole-net, best ep 30/29. *[No (val crops only)]*

2026-10-15 — Stage-2 seeds 1-2 start

src/training/phase5_train_stage2.py --ratio 1 --seed 1/2 → paper_results/stage2/s2_ratio1_s{1,2}, logs/phase5_s2_seeds.log. *[No (train/val crops only)]*

2026-10-15 — Stage-2 seeds 1-2 end

ratio1: s1 AUC 0.9972 F1 0.945 (best ep 30), s2 AUC 0.9962 F1 0.9356 (best ep 16). *[No (val crops only)]*

2026-10-15 — Phase 5 pipeline dry run (val only)

src/pipeline/phase5_pipeline.py --val-only: stage1@0.05 fp/img 2.199; chosen P=0.4 tau=0.27 (recall drop 0.0048), two-stage val fp/img 0.525 vs matched-recall arm B (conf 0.125) 0.967; arm A (0.02) P 0.322 fp/img 4.735 (*[No (val only)]*

2026-10-15 — **Phase 5 official test eval (seed0)**

src/pipeline/phase5_pipeline.py (val choice frozen: P=0.4, tau=0.27, armA 0.02, armB 0.125) → paper_results/twostage/s0/test_report.json. Paired bootstrap C–B: precision +0.128 CI [0.065,0.186]; fp/img –0.253 CI [–0.460, *[Test touch #6 (sanctioned Phase 5 test eval; thresholds from val only)]*

2026-10-15 — Phase 5 pipeline seeds 1-2 end

twostage/s1, s2: all three seeds choose P=0.4 on val (tau 0.27/0.29/0.38); test C precision 0.894/0.859/0.884, fp/img 0.153/0.212/0.169 vs matched-recall B 0.762/0.407. Consistent across seeds. *[Test touches #7/#8 (plan-sanctioned Phase 5 evals)]*

2026-10-15 — Phase 6 start

Plan requires starting from jetson/gradcam.py — file does not exist in repo (jetson/ has build_engine.sh, infer_trt.py, README only). Implementing Grad-CAM fresh per plan §6.0 spec (YOLO: layer feeding detect at stri *[No]*

2026-10-15 — Phase 6 E1/E2/E4 run (after CAM fix)

src/gradcam/phase6_gradcam.py -> paper_results/gradcam/. Fix history: (a) Ultralytics freezes params -> requires_grad_(True); (b) top pre-NMS anchor usually sits on stride-8 head so stride-32 grads were exactly zero -> C *[Test touch #9 (Grad-CAM experiments run on test set, plan-sanctioned Phase 6)]*

2026-10-15 — Phase 6 E6 head-to-head

src/gradcam/phase6_e6_headtohead.py -> e6_pairs.csv, e6_crops.csv, e6_summary.json, e6_grid.png. Both-positive images 182; enrichment median YOLO 85.98 vs MNv3 baseline 1.82; paired diff +147.0 CI [125.4, 170.0] (exclude *[Test touch #10 (Phase 6, plan-sanctioned)]*

2026-10-15 — Phase 6 E5 start

src/gradcam/phase6_e5_flags.py (train+test, conf 0.05) -> logs/phase6_e5.log; lowest-10%-energy flags + worst montages for hand review. *[Test touch #11 (read-only Grad-CAM pass, plan-sanctioned Phase 6)]*

2026-10-15 — Phase 6 E5 complete (fix history)

First run failed: YOLO predict() runs inside torch.inference_mode(), which (a) made the Detect hook crash on retain_grad (guarded) and (b) left anchors/strides as inference tensors + params with requires_grad=False -> Yo *[Test touch #11 (same sanctioned E5 pass; rerun overwrote the aborted attempt)]*

2026-10-15 — Label fixes + with/without

LABEL_FIXES.csv created with 2 objectively-wrong test boxes (ds0_0801 box on tree trunk, ds0_0883 box on fence; E4 review + E5 energy 0.000/nmatch 0). src/evaluation/label_fix_report.py -> paper_results/label_fix_report. *[No new test eval (recomputed from the already-saved raw preds via build_records)]*

2026-10-15 — Phase 6 E3 paste/remove

src/gradcam/phase6_e3_paste.py -> e3_summary.json, e3_paste.csv, e3_remove.csv, example sheets + e3_marker_check.png. Seed 0, 100 composites per condition, deploy conf 0.02. Paste-hold detection: 0.650 @conf0.02, 0.630 @ *[Test touch #12 (plan-sanctioned Phase 6 E3)]*

2026-10-15 — Per-source test results

src/evaluation/per_source_report.py (reads stage2 raw preds + split_manifest source map) -> paper_results/yolo/s_seed0/per_source.{json,md}. Deployment arm A: Jan_lwir 30.100 FP/img (301 of 331 total FPs from 10 images), *[No new model run (recompute from raw preds)]*

2026-10-15 — Rule-11 raw outputs for stage 2

src/analysis/dump_raw_stage2.py -> paper_results/stage2/s2_ratio1_s0/raw_preds_{val,test}.json (conf 0.02, boxes + YOLO conf + cls_p + gt). MNv3 baseline already writes per_image_{val,test}.csv. YOLO raw preds at conf 0. *[Reads test images (no new eval)]*

2026-10-15 — Tables + figures + environment

src/reporting/make_tables.py now emits t1-t6 (.md/.tex) + figures/ (pipeline.png, pr_curves.png, copies of e1_hist/e6_grid/e4 sheet/e5/e3 figures). ENVIRONMENT.txt written (pip freeze, GPU RTX 4070 Laptop 8188 MiB, drive *[No]*

2026-10-15 — Phase 3 seed 1 start

src/training/week3_train_yolov8.py --data <v2 data.yaml> --epochs 75 --batch 8 --name landmine_yolov8s_75ep_s1_v2 --save-period 5 --seed 1 -> logs/phase3_s1_batch8.log (first attempt died pre-flight: missing --data, scri *[No (train/val only; D017 deviation already logged for seed 0)]*

2026-10-15 — Phase 3 seed 1 end + official eval

Training finished 75/75ep (D017 auto-eval in log is not official). src/evaluation/phase3_eval.py --run …s1_v2 --data <v2 root> --out paper_results/yolo/s_seed1. Checkpoint: best @ conf 0.01 (val). Test (frozen protocol *[Test touch #13 (sanctioned Phase 3 seed-1 eval; threshold from val only)]*

2026-10-15 — Phase 3 seed 2 train + official eval

src/training/week3_train_yolov8.py --data <v2 data.yaml> --epochs 75 --batch 8 --name landmine_yolov8s_75ep_s2_v2 --save-period 5 --seed 2 -> logs/phase3_s2_batch8.log (75/75ep). src/evaluation/phase3_eval.py --run …s2_v *[Test touch #14 (sanctioned Phase 3 seed-2 eval; threshold from val only)]*

2026-10-15 — Tables regenerate (3-seed YOLO)

src/reporting/make_tables.py updated: T2 now mean±std over seeds 0/1/2 (val mAP50 0.9544±0.0027, test 0.9768±0.0034, R@OP 0.9763±0.0040, P 0.516±0.218, FP/img 1.510±1.009, OP confs 0.02/0.01/0.21); T3 YOLO deploy/op95 ro *[No (recompute from saved per-seed eval JSONs)]*

2026-10-15 — Test visualization pass (3 seeds)

src/reporting/make_test_vis.py — ultralytics val on val (conf 0.25) + test (frozen OP conf per seed) with save/plots → paper_results/vis/s_seed{N}/{val,test}/vis/. Images only for PDF reports; official metrics unchanged *[Visualization-only (no new threshold/model decision)]*

2026-10-15 — Models release tree

Copied best.pt/last.pt/best.onnx + args.yaml + results.csv for yolo seeds 0/1/2; stage2 best.pt ×3; original_v1 best/last.pt → models/. SHA256SUMS.txt + models/README.md written. .gitignore: !models//*.pt, !models//* *[No]*

2026-10-15 — PDF reports

src/reporting/make_report_pdfs.py (reportlab 5.0.1) → reports/yolov8s_seed{0,1,2}_report.pdf (metrics table, training curves, confusion, PR/P/R/F1, val+test GT/pred batches, aug sample) + reports/research_summary.pdf (he *[No (recompute from saved eval JSONs)]*

2026-10-15 — Master report PDF

src/reporting/make_master_report.py -> reports/master_report.pdf (23 pp): scope/protocol, environment, full dataset narrative (v1 fail, v2 method, freeze), all D001-D023 decisions, Phase 0-6 technical detail, 3-seed YOLO *[No (recompute from saved artifacts)]*

2026-10-24 — Guardrails for extended analysis

scripts/phase2_second_filter.py gained --out and --count-only (sensitivity sweeps no longer overwrite the frozen CSVs); src/pipeline/phase5_build_crops.py gained --pad so context-margin ablations are reproducib *[No]*

2026-10-24 — Tier-1 report corrections

src/reporting/make_master_report.py: filled every placeholder (E1/E2/E5/E6/E3 nested JSON keys; 61 zero-energy TPs; 19/184 and 389/3887; 86.0x vs 1.82x; E3 0.65/0.63/1.00), corrected MNv3 family to Large (code uses mob *[No]*

2026-10-24 — Leaky-v1 train (Block A)

src/training/week3_train_yolov8.py --data <v1 data.yaml> --model yolov8s --epochs 75 --batch 8 --seed 0 --save-period 5 --name landmine_yolov8s_75ep_leakyv1 -> 75/75 ep, 1h 24m 58s. D017 auto test pass at conf 0.25 (v1 *[Test touch #15a (D017 auto pass on v1 test; no decision made from it)]*

2026-10-24 — Leaky-v1 official eval (v1 split)

src/evaluation/phase3_eval.py --run .../landmine_yolov8s_75ep_leakyv1 --data <v1 root> --out paper_results/leaky_v1 -> last.pt, OP conf 0.065 chosen on v1 val. v1 test mAP50 0.9697 / mAP50-95 0.6436; P 0.834 R 0.975; * *[Test touch #15 (sanctioned eval of the leaky baseline)]*

2026-10-24 — Leaky-v1 vs clean-v2 comparison

src/analysis/make_leaky_vs_clean.py: same leaky checkpoint scored on v2 clean test (mAP50 0.9940) and both leaky + clean-v2 seed-0 models scored on the contamination-free v2 test subset (49 imgs). 140/189 (74.1%) of v2 *[Test touch #16 (v2 clean test + clean-subset eval, thresholds frozen from each model's val)]*

2026-10-24 — YOLOv8n seed-0 train + official eval (Block D)

src/training/week3_train_yolov8.py --model yolov8n --epochs 75 --batch 8 --seed 0 --name landmine_yolov8n_75ep_s0_v2 -> 75/75 ep, 2104 s. src/evaluation/phase3_eval.py --out paper_results/yolo/n_seed0: test mAP50 0.9 *[Test touch #17 (size ablation, val-chosen conf)]*

2026-10-24 — **Correction: detector-aligned 3-seed two-stage**

Found that paper_results/twostage/s1 and s2 had been produced with the seed-0 detector (arms A/B byte-identical across seeds). Re-ran src/pipeline/phase5_pipeline.py --yolo <seed-k best.pt> --cls <seed-k best.pt *[Test touch #18 (re-score with frozen configs; no selection)]*

2026-10-24 — Stage-2 ablations (Block B)

src/analysis/make_ablation_matrix.py -> 5 variants (pad 0.0/0.4, crop 160/320, no hard negatives), each: crops (epoch20 FP miner) + 30-ep classifier + raw val/test records. Paired image bootstrap vs baseline: nohardneg *[No new test touch for selection (thresholds chosen on val; test scored once per variant)]*

2026-10-24 — Statistical analysis (Block B)

src/analysis/analysis_stats.py -> paper_results/analysis/{stats.json,stats.md}: image bootstrap (B=2000) per seed, source-cluster bootstrap + leave-one-source-out (3 test sources), shift table, val-chosen policy rule *[No (recompute from saved records)]*

15. Limitations and what was NOT run

• Headline detector results are yolov8s. yolov8n is a single-seed size point on the same frozen split v2. yolov8m was trained only on the older leaky v1 split, so it cannot be compared against the v2 numbers and is deliberately absent from the size sweep.

• Jetson / T7 edge latency (phases 7/8): dropped under approved scope; all numbers are PC GPU FP32.

• Optional retrain on fixed train labels: not run (no objective train-label errors found).

• Dataset itself is not in git (size); the v2 split manifest sha256 is committed.

• Jan LWIR files are colour RGB with marker pins (not thermal); kept with disclosure (D012/D016).

• Stage-2 crop images stay local (regenerable via src/pipeline/phase5_build_crops.py).

• The two-stage gain is detector-dependent: seeds 0 and 1 gain 0.13–0.14 precision over the matched-recall baseline, seed 2 gains nothing (ΔP −0.037, CI spans 0) and is worse on FP/img. With each detector's own detector re-run, the effect is not a robust property of the architecture.

• A val-tuned single-stage threshold is competitive with (and under max-F2 better than) the two-stage arm at moderate recall; the classifier's defensible role is confined to the strict-recall regime (recall ≥ 0.96) and to terrain-nuisance rejection.

• MNv3 baseline is weak (7 negative test images → high seed variance); it is a floor, not a competitor.

• E3 paste-hold uses synthetic composites (feathered edges, scale mismatch) — not a perfect counterfactual.

• Single class (Landmine); no mine-type discrimination.

• Test set has only 5 negative images, so precision and FP/img are high-variance; CIs are reported.

16. Ethics, dual-use, provenance, and availability

Dual use. This work detects visible landmine-like objects and surface indicators from RGB imagery. The same capability could assist clearance operations or be misused for placement/marking avoidance. We report standard detection metrics only, publish no targeting guidance, and frame all claims as a computer-vision benchmark on a public Roboflow-derived dataset — not an operational clearance system. Physical limits (an RGB camera cannot see buried objects) are stated in §1.

Provenance. Images derive from public Roboflow landmine datasets (ds0, ds1) plus May/Jan LWIR-named RGB collections; the original combined dataset and v1 runs are unmodified (D001). The rebuilt v2 split is frozen by a committed manifest SHA-256; the leakage audit, per-image source labels, and all removed-leak candidates are committed as CSV artifacts.

Availability. Code, per-seed weights (best/last/ONNX), tables, figures, and PDFs are in this repository; the dataset itself is not redistributed (size). No license or terms-of-use artifact is included in this release.

17. Extended experiments (30 h budget)

17.1 Leaky-v1 vs clean-v2 — what the old split can and cannot measure

model	eval set	mAP50	P	R	FP/img
leaky v1	v1 test (own split)	0.9697	0.834	0.975	n/a (0 neg)
leaky v1	v2 test, full (contaminated)	0.9940	0.857	0.996	0.222
clean v2 seed 0	v2 test, full	0.9741	0.427	0.976	1.751
leaky v1	v2 test, clean subset (n=49)	0.9950	0.863	1.000	0.204
clean v2 seed 0	v2 test, clean subset (n=49)	0.9947	0.354	1.000	2.347

The leaky model was trained on v1 train and its checkpoint/conf were chosen on v1 val. **140 of 189 v2 test images (74.1%) share a v2 group with an image it trained on**, so its full-v2 score (0.9940) is not a clean generalization number — it is higher than the clean-trained model's 0.9741 precisely because of that contamination. On the 49 contamination-free images the two models tie at 0.9950 vs 0.9947 mAP50 (mAP50-95 0.7250 vs 0.7010; 63 instances, low power). Operating confs differ (0.065 vs 0.02), so P/FP columns are not threshold-matched; mAP is the fair comparison. Reading: the old split did not inflate the model's capability — it made evaluation invalid, model selection leaky, and false-positive behaviour unmeasurable (v1 test has zero negatives). That is why the v2 re-split required retraining.

17.2 MSE threshold sensitivity of the de-duplication filter

MSE ≤	flagged	of 30 leaks captured	same-scene survivors flagged
250	18	18	0
500	25	25	0
750	29	29	0
900	30	30	0
1100	38	30	8

MSE ≤	flagged	of 30 leaks captured	same-scene survivors flagged
1300	51	30	21
1500	65	30	35
2000	95	30	65

All 30 removed leaks have MSE ≤ 817.85 and the nearest surviving candidate is 958.6 — the frozen 900 cut sits in a 6.5% gap. Lowering to 500 would capture only 25/30 leaks; raising to 1500 would additionally flag 35 surviving candidates that are same-scene different frames, not duplicates. The choice is documented in paper_results/analysis/mse_sensitivity.md.

17.3 Detector size points: mAP50 saturates, the operating point does not

run	mAP50	mAP50-95	OP conf	P @ OP	R @ OP	FP/img @ OP
YOLOv8s seed 0	0.9741	0.6924	0.020	0.427	0.976	1.75
YOLOv8s seed 1	0.9807	0.6946	0.010	0.356	0.980	2.38
YOLOv8s seed 2	0.9756	0.6932	0.210	0.764	0.972	0.40
YOLOv8n seed 0	0.9787	0.6892	0.005	0.169	0.984	6.50

The recall-first operating point is the one that matters for a tripwire: it is the strictest YOLO confidence that still reaches 97% recall, chosen on val. YOLOv8n reaches essentially the same mAP50 as the small model but needs conf 0.005 to hit that recall, which leaves 6.5 false detections per test image (vs 1.75 for s seed 0 and 0.15 for the two-stage arm). mAP50 at conf 0.001 therefore hides a large operational difference; every model-size comparison in this report is made at its own val-chosen operating point as well as at mAP. Only v2-trained checkpoints appear here: the medium model was never retrained on v2, so no YOLOv8m row is shown and no statement about medium-size behaviour on the clean split is made.

17.4 Statistical robustness and the honest size of the two-stage gain

All intervals below resample the 189 test images (B = 2000) because the unit of deployment is an image, not a box. Box-level intervals are narrower and would overstate confidence.

run	conf	P	R	FP/img	image recall [95% CI]
seed0	0.020	0.427	0.976	1.751	1.000 [0.980, 1.000]
seed1	0.010	0.356	0.980	2.376	0.995 [0.970, 0.999]
seed2	0.210	0.764	0.972	0.402	0.984 [0.953, 0.994]

Source-cluster bootstrap (three test sources, resampled as clusters) is much wider: precision 0.20–0.87, FP/img 0.14–30.1. The test set is 151 ds0, 28 ds1 and 10 Jan_lwir images, so its source mix — not model variance — dominates the precision estimate. Leave-one-source-out confirms it: removing ds0 gives precision 0.246 at 8.16 FP/img, while removing Jan_lwir gives 0.853 at 0.17. Any single-number test precision should be read with that spread in mind.

Operating-policy rules, all chosen on val and applied unchanged to test

policy	param	val P	val R	test P	test R	test FP/img
recall-safe (primary)	0.270	0.809	0.959	0.894	0.964	0.153
no recall protection	0.020	0.583	0.959	0.654	0.964	0.683
max F2	0.860	0.894	0.950	0.945	0.957	0.074
conf-only above P (classifier off)	1.000	0.902	0.945	0.953	0.957	0.063
YOLO-only: recall-safe	0.045	0.480	0.967	0.576	0.976	0.963
YOLO-only: matched recall to two-stage	0.195	0.771	0.955	0.825	0.968	0.275
YOLO-only: matched FP/img to two-stage	0.225	0.808	0.952	0.854	0.968	0.222
YOLO-only: max F2	0.520	0.956	0.936	0.976	0.949	0.032

Seed 0, the same checkpoint the headline uses. Three readings matter. First, the recall-protection rule is what produces the headline: with the classifier allowed to veto any proposal (no protection) precision falls to 0.654 at the same recall, and protection alone (classifier off, conf ≥ 0.4) already gives P 0.953 / R 0.957 / 0.063 FP/img. Second, at matched FP budget (0.225 conf) YOLO-only gives P 0.854 / R 0.968 / 0.222 FP/img versus 0.894 / 0.964 / 0.153 for the two-stage arm: a real but modest gain in the high-recall regime. Third, if the operating rule is max-F2 instead of recall-safe, a single-stage threshold at conf 0.52 wins (P 0.976 / R 0.949 / 0.032). The two-stage arm is therefore justified by the strict-recall regime, not as a general-purpose improvement, and the paper must say so.

Seed dependence of the primary rule

seed	C P	C R	C FP/img	B P	B FP/img	ΔP (C − B) [95% CI]
0	0.894	0.964	0.153	0.762	0.407	+0.128 [+0.065, +0.186]
1	0.825	0.968	0.275	0.677	0.624	+0.144 [+0.096, +0.181]
2	0.724	0.972	0.497	0.764	0.402	-0.037 [-0.077, +0.013]

Each seed's detector is different, so B (YOLO-only, threshold chosen on val to match C's recall) is the fair per-seed baseline. Seeds 0 and 1 gain 0.13–0.14 precision with a paired interval excluding zero; seed 2 gains nothing (ΔP −0.037, CI spans 0) and is actually

worse than its own B on FP/img (0.497 vs 0.402). The mean ΔP is +0.080 but the across-seed sd is 0.104, so the effect is best described as detector-dependent rather than a robust property of the two-stage design.

Stage-2 ablations (paired against the frozen baseline on the same images)

variant	pad	crop	hardneg	cls thr	P	R	FP/img	ΔP vs base [95% CI]
baseline	0.2	224	True	0.27	0.894	0.964	0.153	reference
pad0	0.0	224	True	0.16	0.866	0.968	0.201	-0.027 [-0.043, -0.011]
pad04	0.4	224	True	0.25	0.868	0.964	0.196	-0.025 [-0.042, -0.011]
crop160	0.2	160	True	0.43	0.911	0.972	0.127	+0.017 [+0.000, +0.036]
crop320	0.2	320	True	0.08	0.868	0.964	0.196	-0.025 [-0.049, -0.006]
nohardneg	0.2	224	False	0.35	0.837	0.972	0.254	-0.056 [-0.084, -0.028]

Removing hard-negative mining is the only clearly harmful change (ΔP −0.056, CI [−0.084, −0.028]): the classifier is mostly a terrain-nuisance rejector, which is consistent with the Grad-CAM dependence result. Both smaller and larger context hurt slightly (pad 0.0 and 0.4, ΔP −0.027 and −0.025), so the 20% margin is a shallow but real optimum. Crop resolution is flat within noise: 160 is nominally better but its interval touches zero (+0.017, [0.000, 0.036]) and 320 is worse (−0.025). No claim of a resolution effect is warranted.

Class balance the test set does not show

split/source	images	boxes	boxes/img	empty
test/Jan_lwir	10	77	7.70	0.0%
test/ds0	151	148	0.98	2.0%
test/ds1	28	28	1.00	7.1%
train/Jan_lwir	100	740	7.40	0.0%
train/May_lwir	776	7706	9.93	0.0%
train/ds0	2589	2643	1.02	0.8%
train/ds1	442	461	1.04	0.0%
valid/Jan_lwir	42	287	6.83	0.0%
valid/ds0	138	131	0.95	5.1%
valid/ds1	1	1	1.00	0.0%

Train is dominated by May_lwir (776 images, 9.9 mines per image), which contributes no test images at all; the test set is 80% ds0 with about one mine per image and 2% empty images. Valid mixes 42 Jan_lwir images (6.8 mines each) with 138 ds0 images, so the val operating point is chosen on a materially different source mix from test. This is a dataset limitation, not something a threshold rule can fix, and it is the main reason the precision estimates move so much under leave-one-source-out.

18. Reproduction

Every artifact is regenerable from committed scripts:

```
python scripts/phase1_audit.py --dataset landmine_combined_v2 --out paper_results/audit_v2_final

python src/training/week3_train_yolov8.py --data --epochs 75 --batch 8 --seed {0,1,2} --save-period 5 --name landmine_yolov8s_75ep_s{N}_v2

python src/evaluation/phase3_eval.py --run runs/detect/runs/detect/landmine_yolov8s_75ep_s{N}_v2 --data --out paper_results/yolo/s_seed{N}

python src/baseline/phase4_mnv3_baseline.py --stage AB --imgsz 224 --seed {0,1,2} ...

python src/pipeline/phase5_build_crops.py && python src/training/phase5_train_stage2.py --ratio 1 --seed {0,1,2} && python src/pipeline/phase5_pipeline.py

python src/gradcam/phase6_gradcam.py && python src/gradcam/phase6_e5_flags.py && python src/gradcam/phase6_e6_headtohead.py && python src/gradcam/phase6_e3_paste.py

python src/analysis/make_mse_sensitivity.py && python src/analysis/analysis_stats.py

python src/reporting/make_tables.py && python src/reporting/make_test_vis.py && python src/reporting/make_report_pdfs.py && python src/reporting/make_master_report.py
```

Repository: https://github.com/tejaskunta/Aeroguard — full git history from Phase 0 inventory through this report. Tags: pre-audit-v2, phase1-audit-fail, phase2-v2-freeze.